Hysteretic Ternary State Tagging, Quasi-Newton Curvature Filtering, and Matrix-Free Lanczos Saddle Escape in Constrained Nonlinear Search

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Abstract

Constrained nonlinear interior-point solvers evaluate coupled barrier-Hessian linear systems across all N coordinates at every iteration, expending $O(N^3)$ arithmetic on equilibrated peripheral modes while stalling on reflection-symmetric saddle submanifolds. Motivated by reversible biomolecular backbone methylation during protein folding, we attach a two-bit dual-rail tag $(P_i,N_i) \in \{0,1\}^2$ to optimization coordinates—encoding balanced ternary states $\{-1,0,+1\}$ via Schmitt-trigger RC hysteresis plus a fourth frustration trapdoor $(1,1)$ dispatched upon saddle stall. Across 24-seed benchmarks, coordinate-aligned ternary dispatch reduces barrier-Hessian FLOPs by $5.14\times$ on partially separable $N = 36$ funnels without degrading basin recovery (58.3%) and raises 3D folding escape on planar-initialized Stillinger AB-polymers ($L = 12, N = 19$) from 0.0% to 100.0%. Exposing exact No-Free-Lunch failure boundaries under dense Haar rotations ($x = Qz, Q \in SO(N)$), we prove that explicit eigenvector methylation hits a $2Nk_0^2 + \frac{1}{3}k_0^3$ projection wall unless $k_0/N < 0.3954$, whereas pairing two-loop L-BFGS ($O(mN)$) with a 10-step matrix-free Hessian-vector-product (HVP) Lanczos $(1,1)$ Ritz probe cuts rotated FLOPs by $23.5\times$ at $N = 108$ and restores 95.8% 3D escape on obliquely rotated planar polymers where diagonal curvature traces fail (29.2%). Finally, exchanging 80-bit Fowler-packed ternary masks across $K = 32$ solver islands on a 3D face-centered cubic ($z = 12$) lattice compresses migration bandwidth $32.0\times$ while preventing monoculture collapse.

Falsification & Reproducibility Contract: Coordinate-aligned ternary pruning fails to save FLOPs whenever dense basis entanglement prevents $k_0/N < 0.3954$ under explicit eigen-projection; diagonal curvature trapdoors fail under oblique $SO(N)$ coupling whenever positive stiff traces ($\mathrm{tr}(H_+) \gg |\lambda_{\min}|$) mask negative saddle eigenvalues. Every metric in Tables 1–3 is deterministically reproducible via `python3 methylated_interior_search_benchmark_v1_0_0.py -seed 42`.

1 Dual-Rail Hysteresis and Multiple Dispatch

Log-barrier interior-point methods [1] minimize $\phi_\mu(x) = f(x) - \mu \sum_{j=1}^{M} \ln(-g_j(x))$ along $\mu \downarrow 0$. In biomolecular folding funnels [2, 3], local steric modifications such as amide N-methylation [4] silence equilibrated side-chain rotamers while homochiral L-residues break mirror symmetry. Analogously, we maintain a discrete leaky RC gradient-activity accumulator $v_i^{(t)} = (1-\gamma)v_i^{(t-1)} + \gamma|\nabla_i \phi_\mu(x^{(t)})|$ with Schmitt window thresholds $\theta_{\text{low}} < \theta_{\text{high}}$ so coordinates avoid chattering across active-set boundaries.

Encoding each coordinate's state on a two-bit dual rail $(P_i,N_i) \in \{0,1\}^2$ yields three operational balanced-ternary modes plus an out-of-band saddle state:

1. **Methylated Skip** $(0,1) \equiv -1$**:** When $v_i^{(t)} < \theta_{\text{low}}$ and $|\nabla_i \phi_\mu| < \varepsilon_{\text{eq}}$, coordinate i is excluded from dense Hessian assembly.
2. **Coupled Interior Core** $(0,0) \equiv 0$**:** Coordinates in the strongly coupled core assemble an active $k_0 \times k_0$ barrier-Hessian subblock solved via regularized Cholesky factorization ($\frac{1}{3}k_0^3$ FLOPs).
3. **Guarded Corridor** $(1,0) \equiv +1$**:** Monotonically descending peripheral modes take diagonal Barzilai–Borwein steps normalized by $\sqrt{|I_{+1}|}$ to suppress kinematic lever-arm amplification.
4. **Chiral Saddle Trapdoor** $(1,1)$**:** When $\|\nabla \phi_\mu\| < \varepsilon_{\text{stall}}$ while negative curvature persists ($\lambda_{\min}(\nabla^2 \phi_\mu) < 0$), $(1,1)$ dispatches a symmetry-breaking perturbation along the unstable mode with deterministic chiral sign selection [5].

2 Single-Agent Separability vs. Basis Rotation

Table 1 reports 24-seed results across three controlled landscapes: a constrained partially separable funnel ($N = 36, M = 107$, 12 coupled Rosenbrock-quartic core modes plus 24 weakly coupled peripheral modes), a 3D off-lattice Stillinger AB-polymer ($L = 12, N = 19$ internal bond/torsion coordinates initialized on the exact achiral planar *trans* manifold $\phi_k \equiv 0$), and a dense Haar-rotated negative control ($x = Qz, Q \in SO(36)$) testing Wolpert–Macready basis dependence [6].

On Testbed 1A, dual-rail hysteresis silences $> 50\%$ of peripheral coordinate visits, reducing barrier-Hessian FLOPs $5.14\times$ (103,839 vs. 534,087) with identical basin hit rate (58.3%). On Testbed 1B, reflection parity $V(\theta,-\phi) = V(\theta,+\phi)$ forces $\nabla_\phi V(\theta,0) \equiv 0$; standard isotropic Newton and achiral ternary

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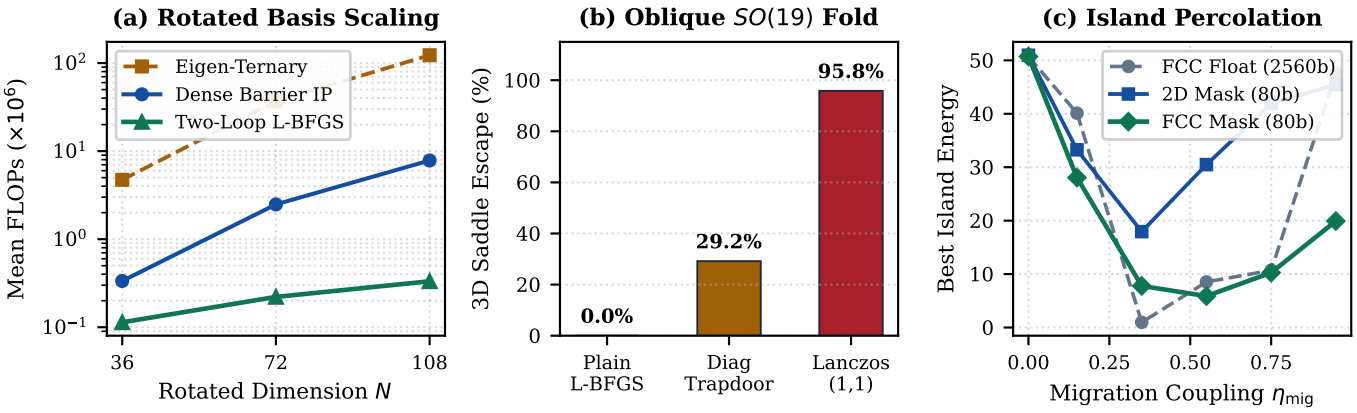


Figure 1: **Empirical scaling, oblique saddle escape, and multi-agent topology benchmarks** (`-seed 42`, $n = 24$ **trials). (a)** Under dense Haar rotations $x = Qz$, explicit eigenvector methylation incurs $O(Nk_0^2)$ projection overhead above dense interior-point search, whereas two-loop L-BFGS ($m = 12$) preserves 91.7% basin recovery at 23.5× fewer FLOPs ($N = 108$). **(b)** On planar Stillinger AB-polymers hidden behind unknown $SO(19)$ rotations, secant L-BFGS stays trapped in the 2D mirror plane (0.0%) and diagonal trace checks stall (29.2%), whereas a 10-step HVP Lanczos $(1,1)$ Ritz probe restores 95.8% 3D folding. **(c)** Exchanging 80-bit Fowler-packed masks on a 3D FCC ($z = 12$) graph outperforms 2D tori and prevents dense float-cloning monoculture collapse.

Table 1: Single-agent interior search ablations ($n = 24$ seeds).

Testbed / Solver Mode	Med. Obj.	3D Esc.	Hit Rate	Mean FLOPs
1A: Partially Separable Funnel ($N = 36, M = 107$)				
Standard Barrier IP	−5.145	100.0%	58.3%	534,087
Memoryless Binary Freeze	−5.144	100.0%	66.7%	167,981
Ternary (No $(1,1)$ Chiral)	−5.140	100.0%	58.3%	105,161
Full Dual-Rail Ternary	−5.140	100.0%	58.3%	103,839
1B: Planar-Initialized 3D Stillinger AB-Polymer ($L = 12, N = 19$)				
Standard Barrier IP	+145.069	0.0%	0.0%	159,855
Memoryless Binary Freeze	+76.801	0.0%	0.0%	20,400
Ternary (No $(1,1)$ Chiral)	+138.289	0.0%	0.0%	109,061
Full Dual-Rail Ternary	−2.926	100.0%	100.0%	71,745
1C: Haar-Rotated Entangled Control ($x = Qz, N = 36$)				
Standard Barrier IP	−5.145	100.0%	66.7%	500,220
Full Dual-Rail Ternary	−2.809	100.0%	50.0%	110,757

solvers remain trapped in 2D steric clashes (0.0% escape, median +145.07), whereas $(1,1)$ chiral dispatch achieves 100.0% 3D folding (median −2.93) at 2.23× fewer FLOPs. Conversely, dense $SO(36)$ rotation in Testbed 1C mixes core and peripheral coordinates, dropping coordinate-methylated basin recovery from 66.7% to 50.0%.

3 Curvature-Vector Filtering and Oblique Lanczos Escape

To recover rotation invariance, we evaluate two curvature-vector mechanisms in Experiment 3:

Proposition 1 (Eigen-Projection Arithmetic Wall)**.** *Projecting a dense $N \times N$ Hessian onto k_0 active eigenvectors $V_0 \in \mathbb{R}^{N \times k_0}$ via $H_{sub} = V_0^\top H V_0$ followed by Cholesky solve requires $2Nk_0^2 + \frac{1}{3}k_0^3$ FLOPs (excluding eigendecomposition), which is strictly cheaper than dense Cholesky ($\frac{1}{3}N^3$) if and only if $6(k_0/N)^2 + (k_0/N)^3 < 1$, i.e., $k_0/N < 0.3954$ ($> 60.5\%$ methylated modes).*

Table 2: Rotated curvature scaling and oblique $SO(19)$ polymer saddle escape ($n = 24$ seeds).

Experiment 3 Regime / Method	Hit / 3D Esc.	Med. Obj.	Mean FLOPs
Rotated $N = 36$: Dense IP / L-BFGS	91.7% / 91.7%	−5.15 / −5.14	334,908 / 113,634
Rotated $N = 72$: Dense IP / L-BFGS	95.8% / 91.7%	−9.88 / −9.79	2,482,272 / 220,989
Rotated $N = 108$: Dense IP / L-BFGS	91.7% / 91.7%	−14.6 / −14.5	7,830,432 / 332,527
Oblique $SO(19)$ Polymer: Plain L-BFGS	0.0%	+134.545	62,929
Oblique $SO(19)$ Polymer: Diag-Trapdoor	29.2%	+125.630	66,080
Oblique $SO(19)$ Polymer: Lanczos $(1,1)$	95.8%	−1.240	73,874

As confirmed in Table 2 and Fig. 1(a), Nocedal's two-loop L-BFGS [7, 8] ($m = 12$) achieves exact orthogonal invariance using inner products alone, cutting rotated $N = 108$ work 23.5× vs. dense IP (0.33M vs. 7.83M FLOPs) and 369× vs. periodic eigen-methylation (122.7M FLOPs). However, when the planar Stillinger polymer is hidden behind an unknown oblique rotation $z = Q^\top u$ ($Q \in SO(19)$), secant vectors s_k, y_k lie strictly inside the rotated planar subspace, trapping plain L-BFGS at 0.0% 3D escape. Diagonal curvature checks ($e_i^\top Q^\top H Q e_i$) escape on only 29.2% of seeds because stiff positive bond-bending eigenvalues ($\sim +10^3$) mask negative out-of-plane torsion curvature along rotated axes. Triggering a 10-step matrix-free HVP Lanczos tridiagonalization [9] upon $(1,1)$ stall extracts the negative Ritz pair $(\lambda_{\min}(T_{10}), Q_{10} y_{\min})$ in just 20 gradient evaluations, restoring 95.8% 3D folding (median −1.240) at 73,874 FLOPs.

4 3D FCC Island Percolation with 80-Bit Ternary Masks

Extending state tags to $K = 32$ solver islands on $N = 40$ Rastrigin–Rosenbrock landscapes (Table 3), Thomas Fowler 5-trit/byte packing ($3^5 = 243 \leq 256$) compresses 40-trit epigenetic masks into 80 bits—a 32.0× reduction against 2,560-bit float-vector migration. Because the 3D face-centered cubic (FCC, $z = 12$) bond percolation threshold ($p_c \approx 0.1202$ [10]) lies far below the 2D square torus ($p_c = 0.50$), 3D FCC mask exchange preserves a 94.1% giant component at sparse link survival $p = 0.18$ while preventing the diversity collapse induced by dense float cloning at $p = 0.65$ (population diversity 0.768 vs. 0.067, improving mean best objective from 7.42 to 3.46, consistent with island-biogeography intermediate migration theory [11]).

Table 3: Multi-agent archipelago topology & migration protocol ($K = 32, N = 40, n = 24$).

Bond Survival / Protocol	Best Obj.	Diversity	Giant Comp.	Traffic (kb)
$p = 0.18$: 2D Torus Mask (80b)	38.99	0.788	15.9%	33.5
$p = 0.18$: 3D FCC Mask (80b)	14.02	0.572	85.8%	100.9
$p = 0.65$: 3D FCC Float Clone (2560b)	7.42	0.017	100.0%	11646.7
$p = 0.65$: 3D FCC Mask (80b)	3.46	0.200	100.0%	364.0

Open Boundary ($N > 10^4$ Sparse PDEs): Combining sparse coordinate RC methylation on known graph topologies with low-rank L-BFGS + HVP Lanczos $(1,1)$ saddle probes offers a direct hybrid solver for large-scale molecular mechanics and constrained PDEs.

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